Mander Portman Woodward

Physics A

Chapter 11
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **34**.
- The total number of marks may take into account some 'either/or' question choices.

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1. Which of the following waves can be polarised and has a typical wavelength of about a few centimetres?

A microwaves
B ultraviolet
C sound
D visible light

Your answer ☐ **[1]**

2. Which of the following statements is/are correct about electromagnetic waves?

1 They can be plane polarised.
2 They can be refracted and diffracted.
3 They have the same speed in a vacuum and in glass.

A Only 1
B Only 3
C Only 1 and 2
D 1, 2 and 3

Your answer ☐ **[1]**

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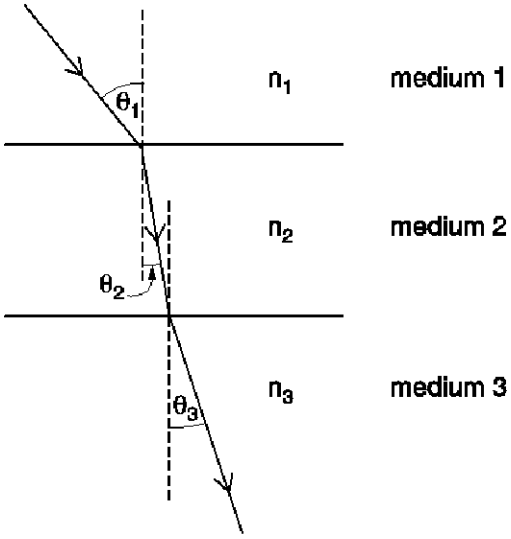
3. A ray of light passes through three media with refractive indices n_1, n_2 and n_3. The speed of light in medium 1 is v_1, in medium 2 is v_2 and in medium 3 is v_3. The angle between the ray and the normal in medium 1 is θ_1, θ_2 in medium 2 and θ_3 in medium 3.



Which of the following statements is/are true?

A. The velocity of light in medium 3 is equal to the velocity of light in medium 1

B. $v_3 = v_1 \frac{\sin\theta_3}{\sin\theta_1}$

C. The frequency of light in medium 2 is less than the frequency in medium 1

D. $\frac{n_2}{n_1} = \frac{v_2}{v_1}$

Your answer ☐

[1]

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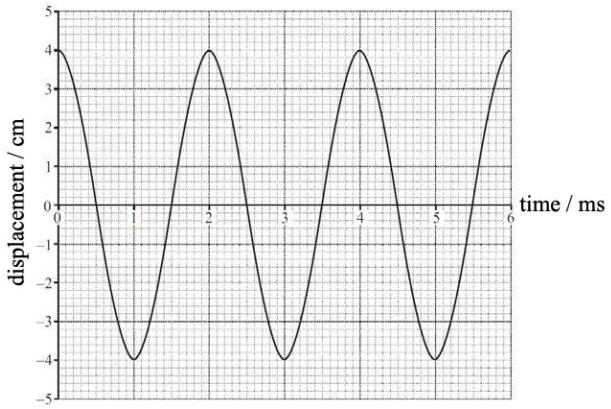
4. The diagram below shows the displacement-time graph of an air particle as a sound wave passes.



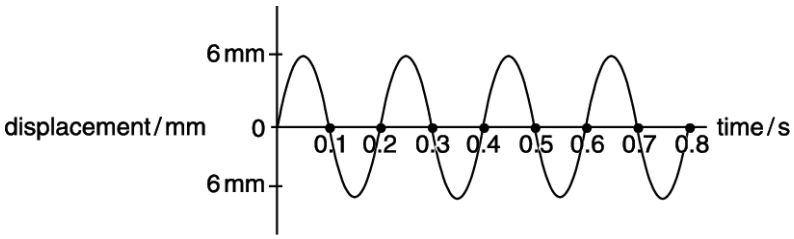
The speed of the sound wave is 340 m s^{-1}.
What is the wavelength of the sound wave?

A. 0.68 m
B. 1.7 m
C. 170 m
D. 680 m

Your answer ☐

[1]

5. The graph shows the variation of displacement with time for a progressive wave.



Which of the following statements can be deduced from the graph?

A. The frequency of the wave is 5 Hz.
B. The graph represents a transverse wave motion.
C. The amplitude of the wave is 12 mm.
D. The wavelength of the wave is 60 m.

Your answer ☐

[1]

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6. A student views the display of a laptop screen through a polarising filter. The intensity of the light changes when the filter is rotated.

Which property of light is demonstrated in this experiment?

A It has wavelength of about 5×10^{-7} m.
B It travels at the speed of light.
C It is a transverse wave.
D It is a longitudinal wave.

Your answer ☐

[1]

7. Which of the following statements is true about electromagnetic radiation?

A. Radio waves are in the highest energy range of the electromagnetic spectrum.
B. Visible light has a range of wavelengths in the order of 450 nm to 700 nm.
C. Ultra-violet waves cannot be plane polarised.
D. Gamma rays have a low frequency.

Your answer ☐

[1]

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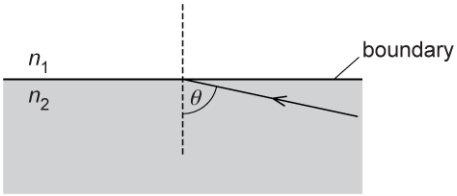
8. A ray of monochromatic light is incident at the boundary between two transparent materials of refractive index n_1 and n_2. The critical angle θ is equal to 80°.



What is the ratio $\frac{n_1}{n_2}$?

A 0.17

B 0.98

C 1.02

D 5.76

Your answer ☐ **[1]**

9. A progressive wave of amplitude *a* has intensity *I*. This wave combines with another wave of amplitude 0.6*a* at a point in space. The phase difference between the waves is 180°.

What is the resultant intensity of the combined waves in terms of *I*?

A 0.16 *I*

B 0.4 *I*

C 1.6 *I*

D 2.6 *I*

Your answer ☐ **[1]**

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10. The table shows the refractive index *n* of four transparent materials **A**, **B**, **C** and **D**.

Which material has the **smallest** critical angle?

Material	**A**	**B**	**C**	**D**
n	2.01	1.87	1.60	1.33

Your answer ☐ **[1]**

11. Describe the difference between longitudinal and transverse waves.

---**[2]**

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12. Fig. 4.1 shows a section of a uniform string under tension at one instant of time. A progressive wave of wavelength 80 cm is moving along the string from left to right. At the instant shown, the displacement of the string is zero at the point opposite the zero mark on the scale beneath the string.

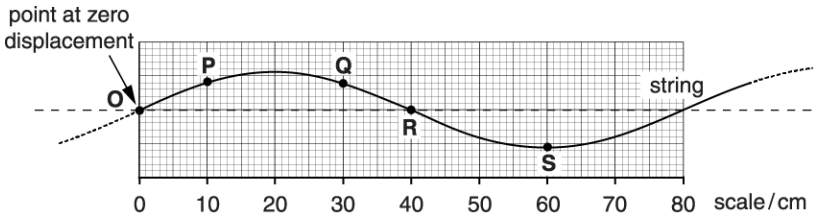

Fig. 4.1

Four points **P**, **Q**, **R** and **S** at 10, 30, 40 and 60 cm respectively, are marked on the string. The oscillatory motion of each point can be described in terms of amplitude, frequency and phase difference from **O**.

i. State the meaning of each of the terms

1. *amplitude*

---

2. *frequency*

---

3. *phase difference*.

---

ii. Describe using these three terms how the motion of points **P**, **Q**, **R** and **S**

1. is similar,

---

2. is different.

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13. Fig.17 shows a ray of light at the boundary between glass and water.

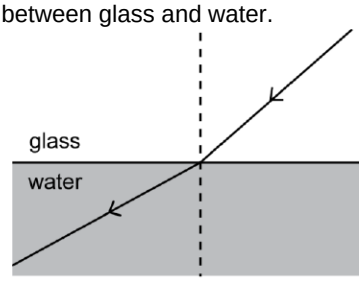

Fig. 17 (not to scale)

Use Fig. 17 to describe and explain how the wavelength of the light changes as light travels from glass to water.

[3]

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14. Fig. 26.1 shows the variation of displacement *y* with position *x* of a progressive transverse wave on a stretched string at a particular instant.

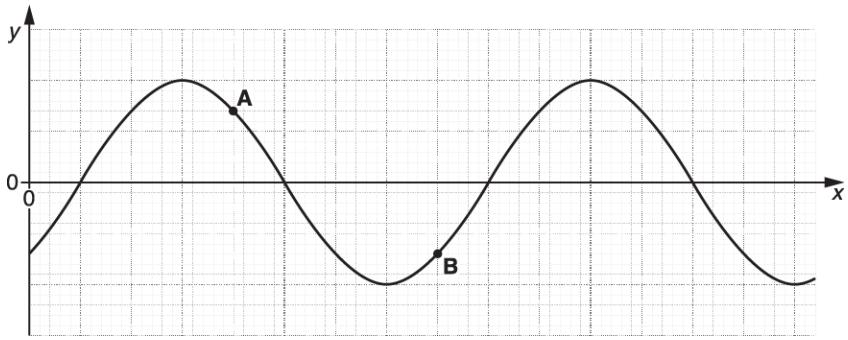

Fig. 26.1

The motions of particles **A** and **B** of the string is analysed over a short period of time. The distance between the positions of **A** and **B** is half a wavelength of the wave. The particles **A** and **B** have the same speed.

i. State **one** difference between the motions of these particles.

.....

..... **[1]**

ii. The particle **A** oscillates with frequency 75 Hz.
The distance between the positions of **A** and **B** is (40.0 ± 2.0) cm.
Calculate the speed *v* of the transverse wave on the string and the absolute uncertainty in this value.

v = ... ± ..... m s^{-1} **[3]**

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15.

In a ripple tank experiment, a dipper vibrates on the surface of water.
Circular waves spread out in all directions from the dipper.
The variation of displacement of the water with distance *x* from the dipper at one instant in time is shown in Fig. 25.1.

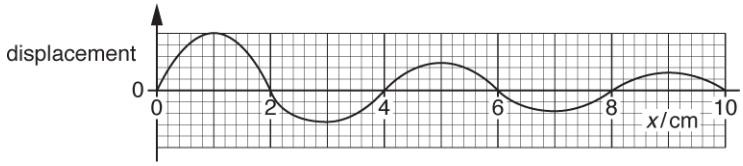

Fig. 25.1

i. Determine the wavelength λ of the wave in cm.

λ = ...

ii. Explain why the **intensity** of the wave changes as the distance *x* increases.

.....

.....

.....

..... **[2]**

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16.

Light travels from air to water. The refractive index of water is greater than the refractive index of air.
Compare the speed, frequency and wavelength of light in air and in water.

---**[3]**

17.

In an experiment using microwaves, a metal grille **G** consisting of a series of long metal rods is placed between a transmitter **T** and a detector **D** as shown in Fig. 7.2.

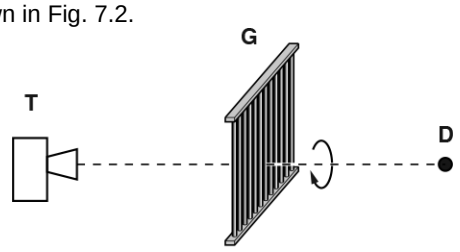

Fig. 7.2

The grille is slowly rotated through 180° about the line joining **T** and **D**. The detected signal at **D** varies from zero to maximum and back to zero again.

Explain why the detected signal behaves in this way.

---**[2]**

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18. A narrow beam of unpolarised light is incident at the boundary between air and glass.

Fig. 18 shows the incident ray, the reflected ray and the refracted ray at the air-glass boundary.

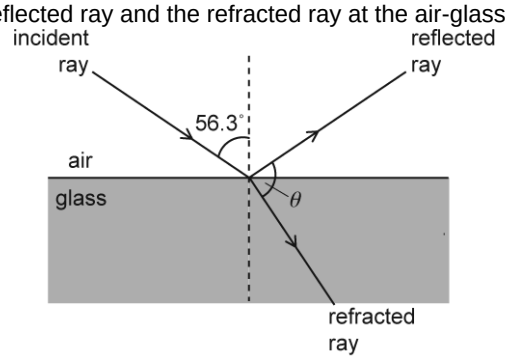

Fig. 18 (not to scale)

The refractive index of air is 1.00 and the refractive index of the glass is 1.50.
The angle of incidence of the light is 56.3°.

Describe how you can demonstrate in the laboratory that reflected light is plane polarised.

--- **[2]**

END OF QUESTION PAPER

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Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			**A**	1	
			Total	**1**	
2			C	1	
			Total	**1**	
3			B	1	
			Total	**1**	
4			A	1	
			Total	**1**	
5			A	1	
			Total	**1**	
6			C	1	**Examiner's Comments** All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions. The candidates to demonstrate their knowledge and understanding of physics.
			Total	**1**	
7			B	1	
			Total	**1**	
8			**B**	**1**	
			Total	**1**	
9			**A**	**1**	
			Total	**1**	
10			**A**	1	**Examiner's Comments** This question was about the critical angle and refractive index equation sinC = 1/*n* which appears in the Data, Formulae and Relationship Booklet. The majority of the candidates realised that the largest value of

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					the refractive index would give the smallest critical angle. The answer had to be A. It was good to see the equation above scribbled on many of the scripts.
			Total	**1**	
11			Transverse: vibrations / oscillations are perpendicular / right angles to the direction of travel / energy transfer (AW)	B1	**Allow** 1 mark for 'For one of the waves, the oscillations / vibrations are at right angles and for the other they are parallel to the direction of travel' (AW)
			Longitudinal: vibrations / oscillations are parallel to / in the same direction as the direction of travel / energy transfer (AW)	B1	**Not** move for vibrations / oscillations **Allow 1** mark for transverse (waves) can be polarised ORA **Examiner's Comments** The opening question was supposed to be accessible to all the candidates. A significant number of candidates did not gain credit because of vague terminology, with 'move' or 'travel' often used to mean two things at once or not giving sensible discrimination between the directions of oscillation and wave / energy travel.
			Total	**2**	
12		i	the maximum displacement from equilibrium or rest position	B1	**allow** *zero* or *undisturbed* for *equilibrium*
		i	number of oscillations / vibrations (at a point) per unit time	B1	number of wavelengths passing a point or produced by the wave source per unit time **allow** per second **NOT** *amount* for *number*
		i	how far 'out of step' (out of sync) the oscillations at two points on the wave / string are / AW	B1	**alt e.g.** the fraction of a cycle between the oscillations at the two points
		ii	all have same frequency **or** same amplitude	B1	**N.B.** withhold mark if extra incorrect answers given
		ii	all have different phases / phase differences	B1	**allow** *not in phase* or *all out of phase* **Examiner's Comments** The basic definitions on waves in part (i) were often imprecise. A typical example for *frequency is the amount of waves in a second. For phase difference* most compared two different waves rather than referring to two points on the same wave. In part (ii) many confused a progressive wave with a stationary wave so only about one third of the candidates scored both marks.
			Total	**5**	
13			The ray is refracted away from the normal, therefore the refractive index of water is less than the refractive index of glass or speed of light in water is	B1	

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			greater than the speed of light in glass.		
			The frequency remains constant.	B1	
			$v = f\lambda$ and therefore the wavelength of light increases as it travels from glass to water.	B1	
			Total	**3**	
14		i	**A** and **B** move in opposite directions	B1	**Allow A** is moving up and **B** is moving down (or vice versa) **Allow** they have a phase difference of 180(°) or π (rad) **Allow** they are in antiphase **Examiner's Comments** The majority of the candidates gave a good answer. Most realised that the particles at **A** and **B** will be moving in opposite directions or have a phase difference of 180°.
		ii	λ = 0.80 (m) $v = f\lambda$, $v = 75 \times 0.80$	C1	**Allow** 80 (cm) for this C1 mark
		ii	$v = 60$ (m s^{-1}) absolute uncertainty = $\frac{2.0}{40} \times 60$	A1	**Allow** 1 mark for 30 (m s^{-1}) from the C1A1 marks; λ = 0.40 m used
		ii	absolute uncertainty = 3.0 (m s^{-1})	A1	**Note** 60 ± 3 (m s^{-1}) scores full marks **Allow** 2 marks for 6000 ± 300 (m s^{-1}); λ in cm (POT error) **Allow** 2 marks for 30 ± 1.5 (m s^{-1}); λ = 0.40 m used **Examiner's Comments** This was a notable success for the candidates; many correctly determined the wave speed to be 60 m s^{-1}. The absolute uncertainty of 3.0 m s^{-1} was correctly calculated by most of the top-end candidates. The most frequent incorrect values for the uncertainty were 0.02 m s^{-1} and 0.04 m s^{-1}. A significant number of the low-scoring candidates took the wavelength to be 0.40 m. This gave an answer of (30 ± 1.5) m s^{-1}. Examiners awarded two marks for such an answer.
			Total	**4**	
15		i	4 (cm)	**B1**	**Examiner's Comments** Almost all scripts had the correct answer of 4.0 cm for the wavelength.
		ii	(As the wave spreads out the) amplitude decreases	**M1**	**Not** 'displacement' **Not** '*A* decreases' **Ignore** 'energy is lost'
		ii	intensity ∝ amplitude^2 **and** therefore intensity decreases	**A1**	**Allow** $I \propto A^2$ **Note** Do not allow this mark if we also have $I \propto 1/x^2$ but **allow** this mark if we also have $I \times 1/x$

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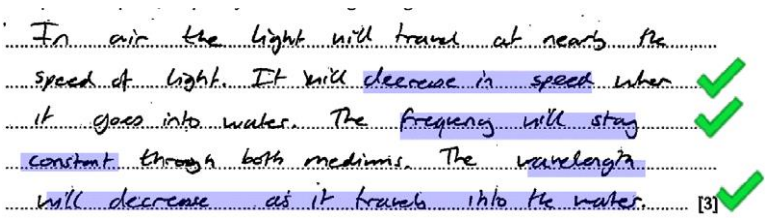
					Allow 1 mark for: **(***I* = *P* / *A)* power is constant and as area increases the intensity decreases **or** intensity ∝ 1/area and as area increases the intensity decreases **Examiner's Comments** About half of the candidates scored one or more marks for their explanations. A fair number of candidates realised that the amplitude decreased and this led to a decrease in intensity because *intensity* ∝ *amplitude*^{2}. No credit was given to candidates who stated that the *'intensity decreased because of the inverse square law'.* The intensity of circular waves spreading out on the surface of water does not obey this law. A significant number of candidates incorrectly stated that the *'intensity and power decreased because of heat losses or friction'.*
			Total	**3**	
16			Speed of light is less in water (ORA) Frequency is the same (in both) Wavelength is smaller in water (ORA)	**B1** **B1** **B1**	**Allow** calculated values for air **and** water **Allow** speed decreases (from air to water) **Not** *v* or *c* **Allow** *f* is the same **Allow** wavelength / *λ* decreases (from air to water) **<u>Examiner's Comments</u>** Most candidates gained two or more marks. Many candidates were aware that the speed of light was less in water than in air. A significant number of candidates also knew that the frequency of light remains constant and successfully argued the fate of the wavelength using the wave equation *v* = *fλ*. **Exemplar 12**  In air the light will travel at nearly the speed of light. It will decrease in speed when it goes into water. The frequency will stay constant through both mediums. The wavelength will decrease as it travels into the water. [3] This exemplar illustrates a flawless answer from a top–end candidate. It had all the main ingredients for scoring 3 marks. The answers matched well with the marking points – the examiner had no issues with following the text.
			Total	**3**	
17			Microwaves from **T** are transverse/polarised wtte	**B1**	**Allow** E field perpendicular to direction of motion

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			At 0° or 180° the grille blocks (all) the (polarised) waves <u>and</u> at 90° the grille allows all the microwaves to pass.	**B1**	**Allow** explanation in terms of $I = I_0 \cos^2\theta$ **<u>Examiner's Comments</u>** Candidates found this question difficult. Candidates often did not state that the microwaves were polarised or mistakenly thought that the grille caused the microwaves to become polarised. Candidates also did not appear to read the question carefully often thinking that the detected signal varied from a maximum initially. Some candidates quoted $I = I_0 \cos^2\theta$ to help explain their answer.
			Total	**2**	
18			Use a polaroid / polarising filter Rotation will change intensity	B1B1	**Allow** brightness / light **<u>Examiner's Comments</u>** Most candidates scored well deserved 2 marks here for opting to use a polarising filter and rotating it to see variation in the intensity of the transmitted light. A small number of candidates chose the wrong equipment – the most common of these were diffraction grating, a single slit and microwave receiver.
			Total	**2**	

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